Prepare for Qualitative Data Collection

Emin Zerman

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Project Work – So Far

- Form groups of 3-5 students.
- Come up with a project idea. What problem or need are you going to solve?
- Think about who your target group is!

Timeline

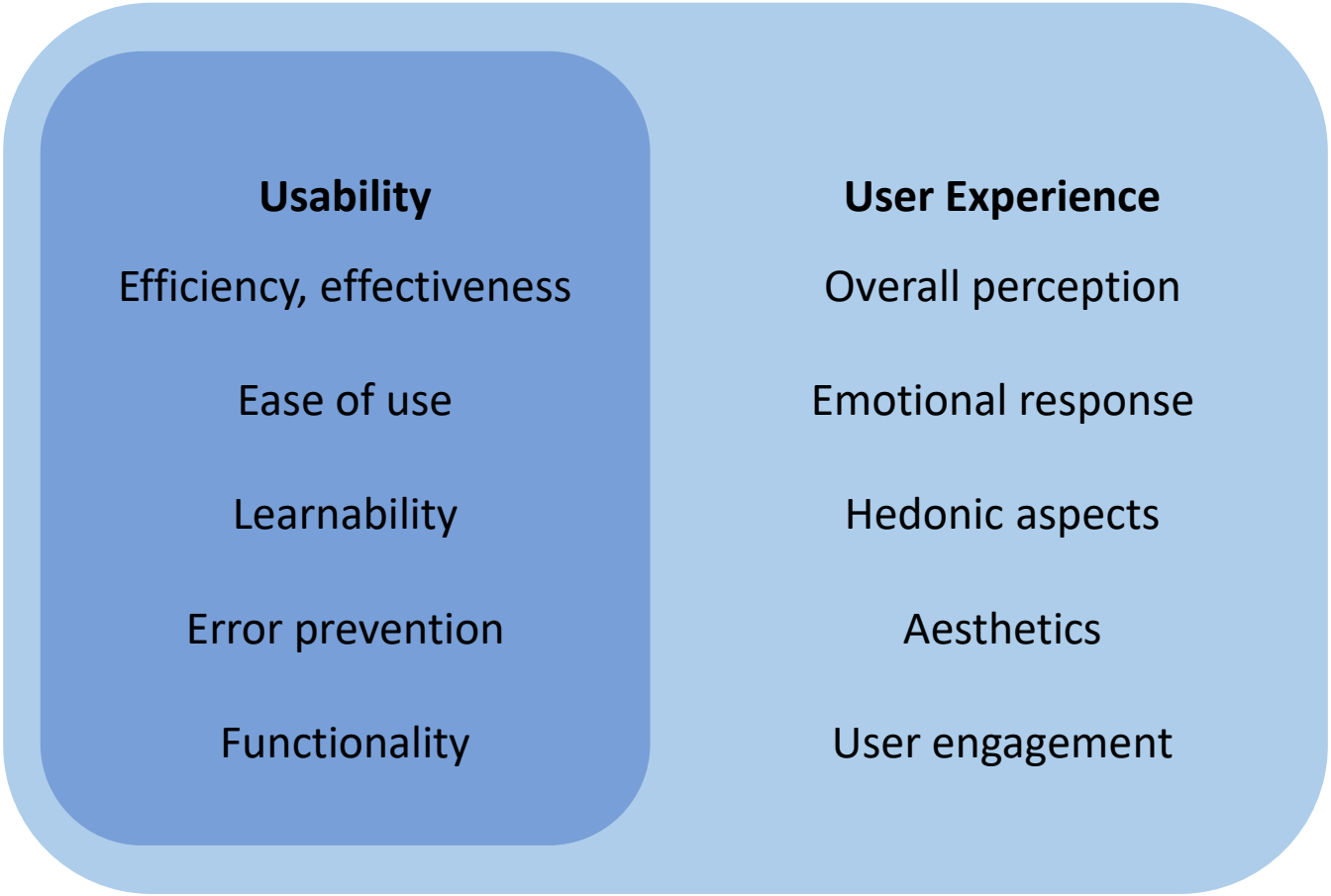
	Mon	Tue	Wed	Thu	Fri
v38	Project start (C015 and C016, ALC) →				Data collection (M209)
v39			Data analysis (M209)		
v40					
v41	Prototypes (C015 and C016, ALC)			Usability tests (M209)	
v42					
v43					
v44			Presentations	Presentations	

Project Work – Throughout the process

- What usability and user experience goals are most important to your target audience in this interface?

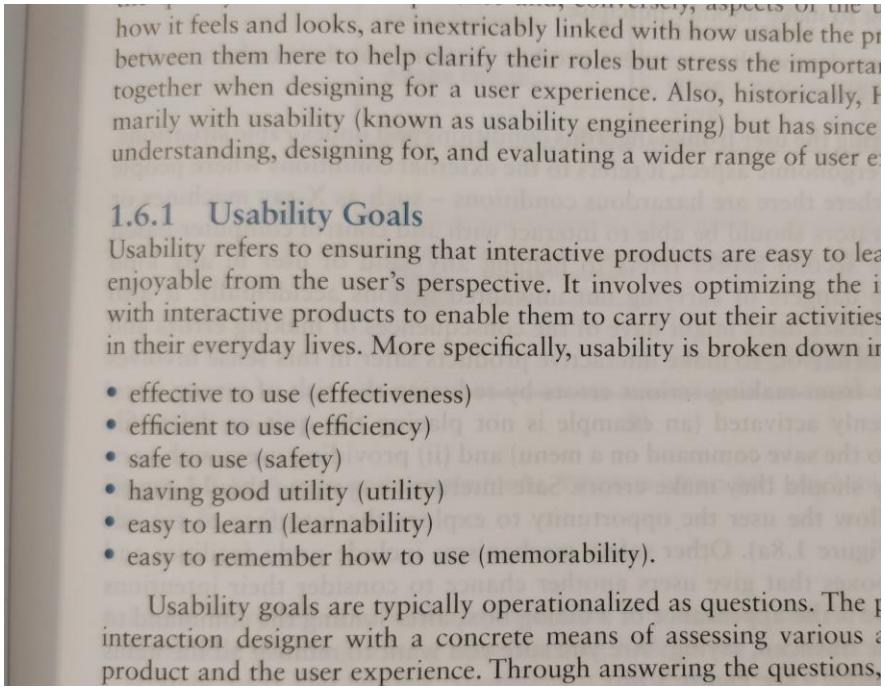
- Look for inspiration in, for example, current apps and products (environmental monitoring)

Repetition – Usability vs. UX



Repetition – Usability vs. UX

Usability goals



how it feels and looks, are inextricably linked with how usable the pr
between them here to help clarify their roles but stress the importa
together when designing for a user experience. Also, historically, H
marily with usability (known as usability engineering) but has since
understanding, designing for, and evaluating a wider range of user e

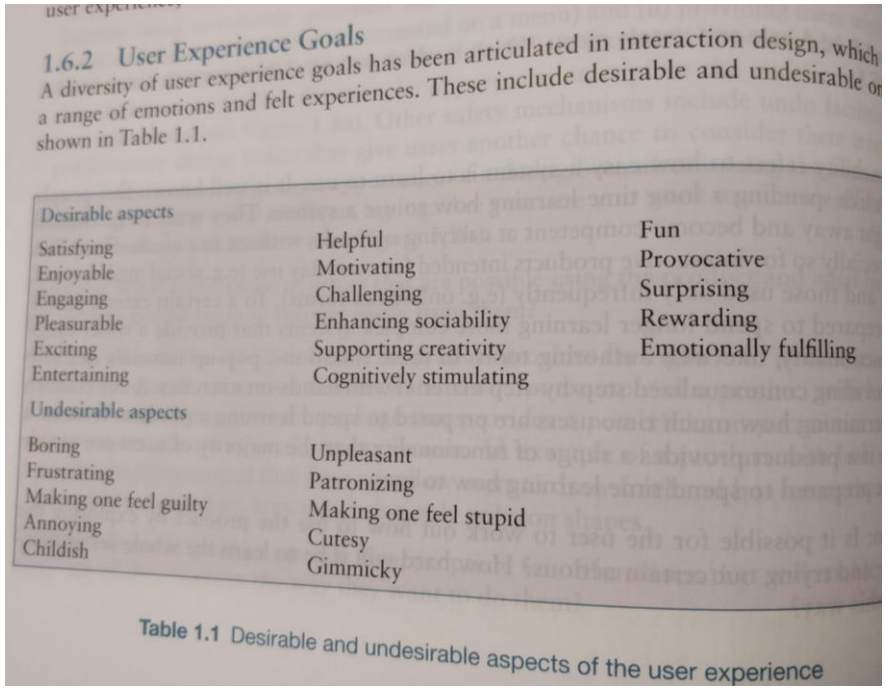
1.6.1 Usability Goals

Usability refers to ensuring that interactive products are easy to lea
enjoyable from the user's perspective. It involves optimizing the i
with interactive products to enable them to carry out their activities
in their everyday lives. More specifically, usability is broken down in

- effective to use (effectiveness)
- efficient to use (efficiency)
- safe to use (safety)
- having good utility (utility)
- easy to learn (learnability)
- easy to remember how to use (memorability).

Usability goals are typically operationalized as questions. The p
interaction designer with a concrete means of assessing various a
product and the user experience. Through answering the questions,

User experience goals



user experience.

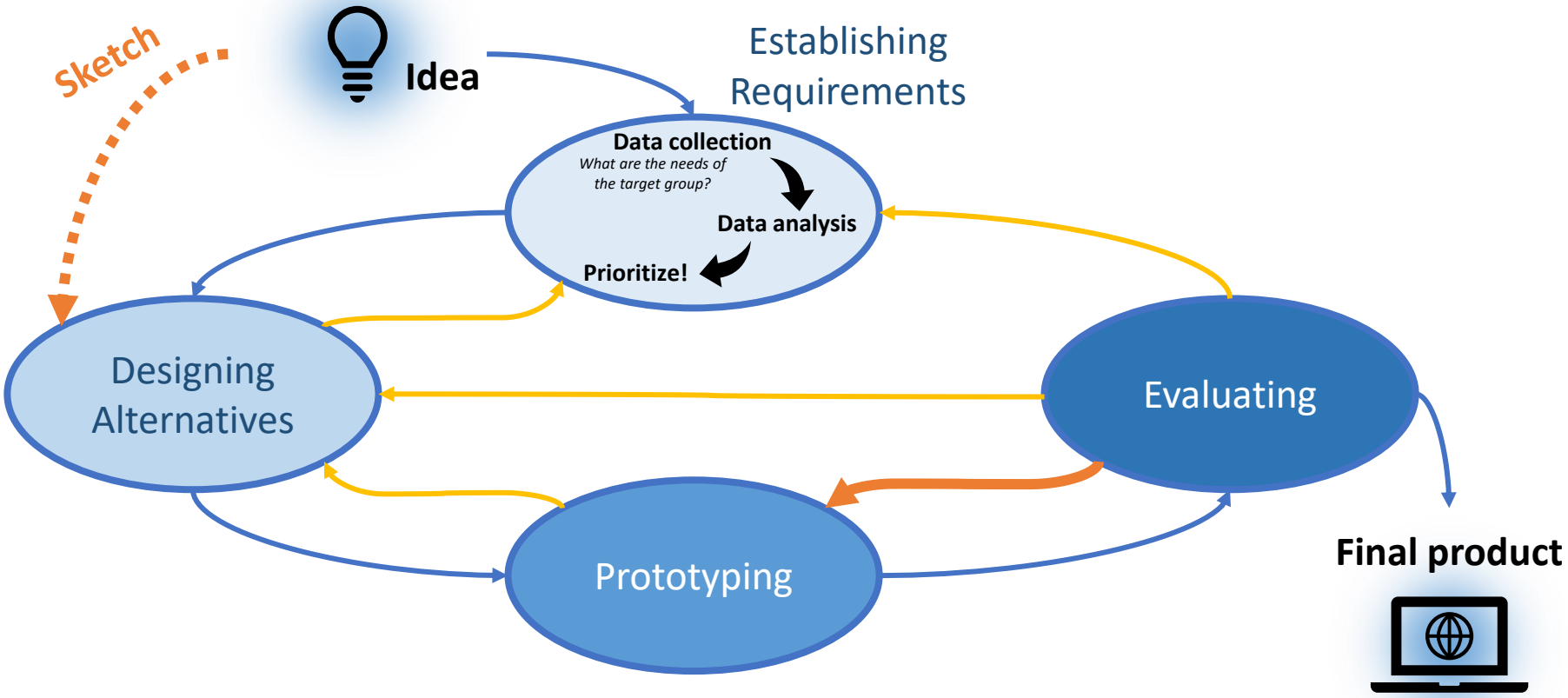
1.6.2 User Experience Goals

A diversity of user experience goals has been articulated in interaction design, which
a range of emotions and felt experiences. These include desirable and undesirable or
shown in Table 1.1.

Desirable aspects		
Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhancing sociability	Rewarding
Exciting	Supporting creativity	Emotionally fulfilling
Entertaining	Cognitively stimulating	
Undesirable aspects		
Boring	Unpleasant	
Frustrating	Patronizing	
Making one feel guilty	Making one feel stupid	
Annoying	Cutesy	
Childish	Gimmicky	

Table 1.1 Desirable and undesirable aspects of the user experience

Repetition – The process of interaction design



*Before you can **develop** the "right" **solution**,*

*you have to **understand** the "right" **problem**.*

How do I find a problem/solution?

1. Identify Your Interests and Passions:

- Introspection and self-assessment – Think deeply about your own interests and passions. What topics excite you? What are your hobbies? What problems or challenges do you find intriguing in those areas?

2. Explore User-Centered Problems:

- Explore real-world problems or challenges that users face within those areas.
  - Researching news articles, academic papers, and industry reports to identify ongoing issues or trends.
  - Conducting surveys, interviews, or user observations to understand specific pain points and user needs.
  - Analyzing existing apps, products, or services to identify areas where improvements can be made.
- Document your findings, including specific user stories, anecdotes, or scenarios that highlight the challenges users encounter.

How do I find a problem/solution?

1. Identify Your Interests and Passions:

- Think deeply about your own interests and passions.

2. Explore User-Centered Problems:

- Explore real-world problems or challenges

3. Brainstorm Solutions:

- Brainstorm potential solutions. Think creatively about how technology and interactive interfaces can address these problems effectively.
  - Generate multiple solution ideas **without judgment**.
  - Consider the **feasibility** of each idea, including technical constraints and resources.
  - **Prioritize** ideas based on their potential impact, user benefit, and alignment with course goals.

- At this stage, ideas can be high-level concepts and don't need to be fully detailed.

Discussion

What are your interests and passions:

- Sports, gaming, musical instruments, health & wellbeing, solving problems, …

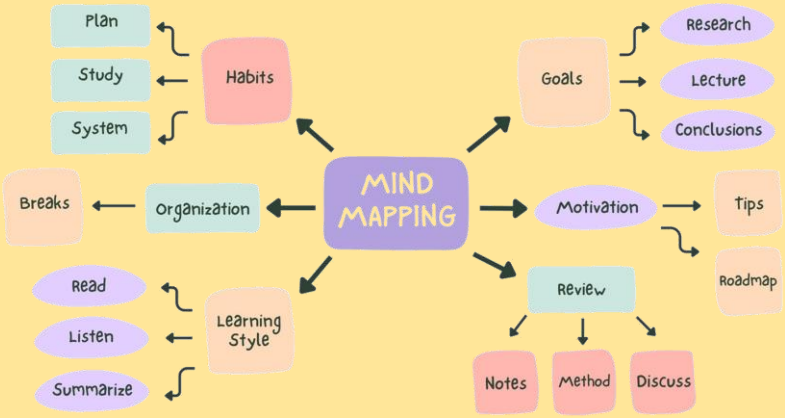
Explore user-centered problems:

- What are the problems that needs to be solved?
  - There are many, hone in on a problem using your interests

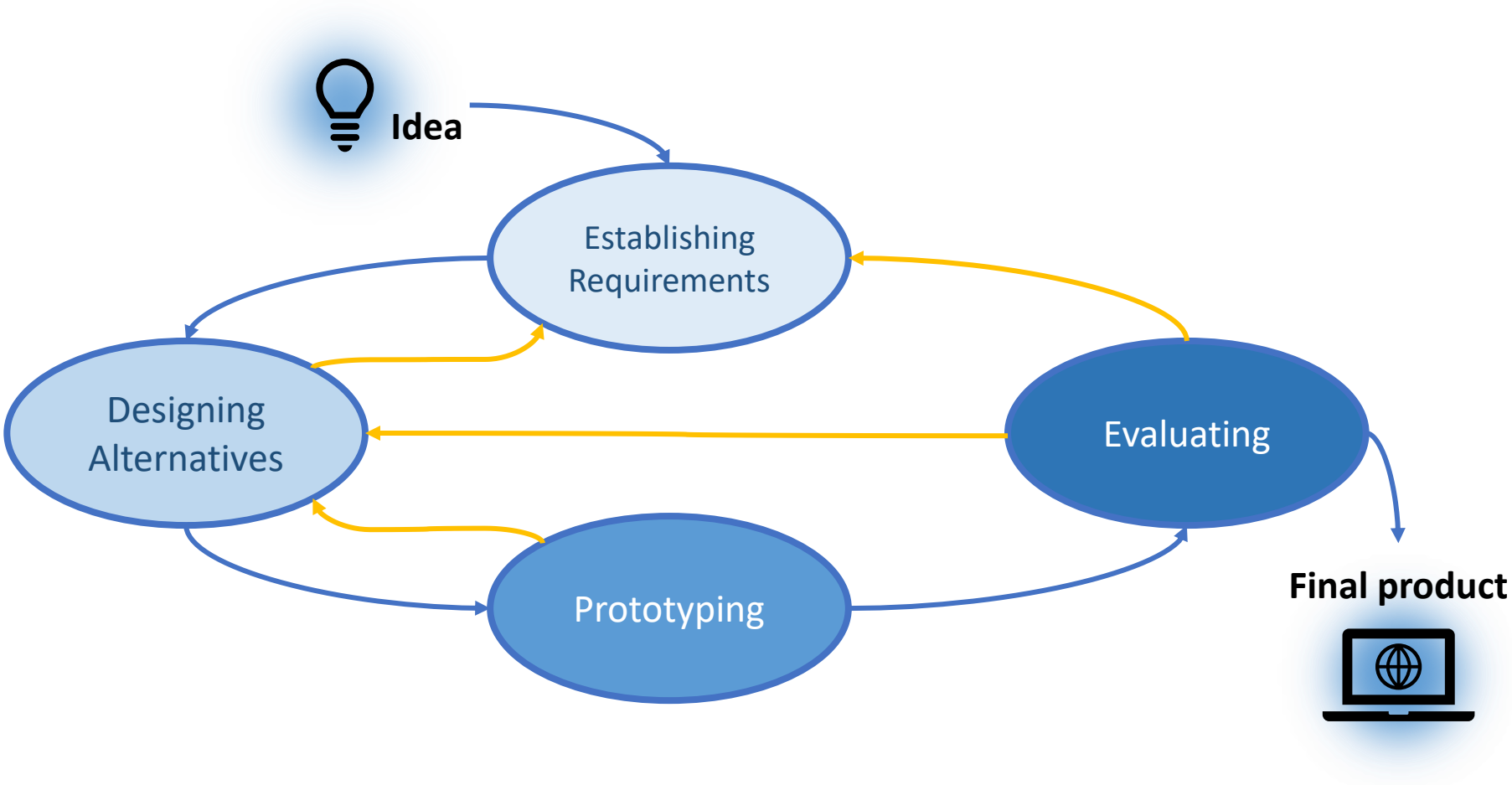
How to brainstorm solutions?

- Mind mapping
- Free writing

  Write without stopping and correcting for a set amount of time

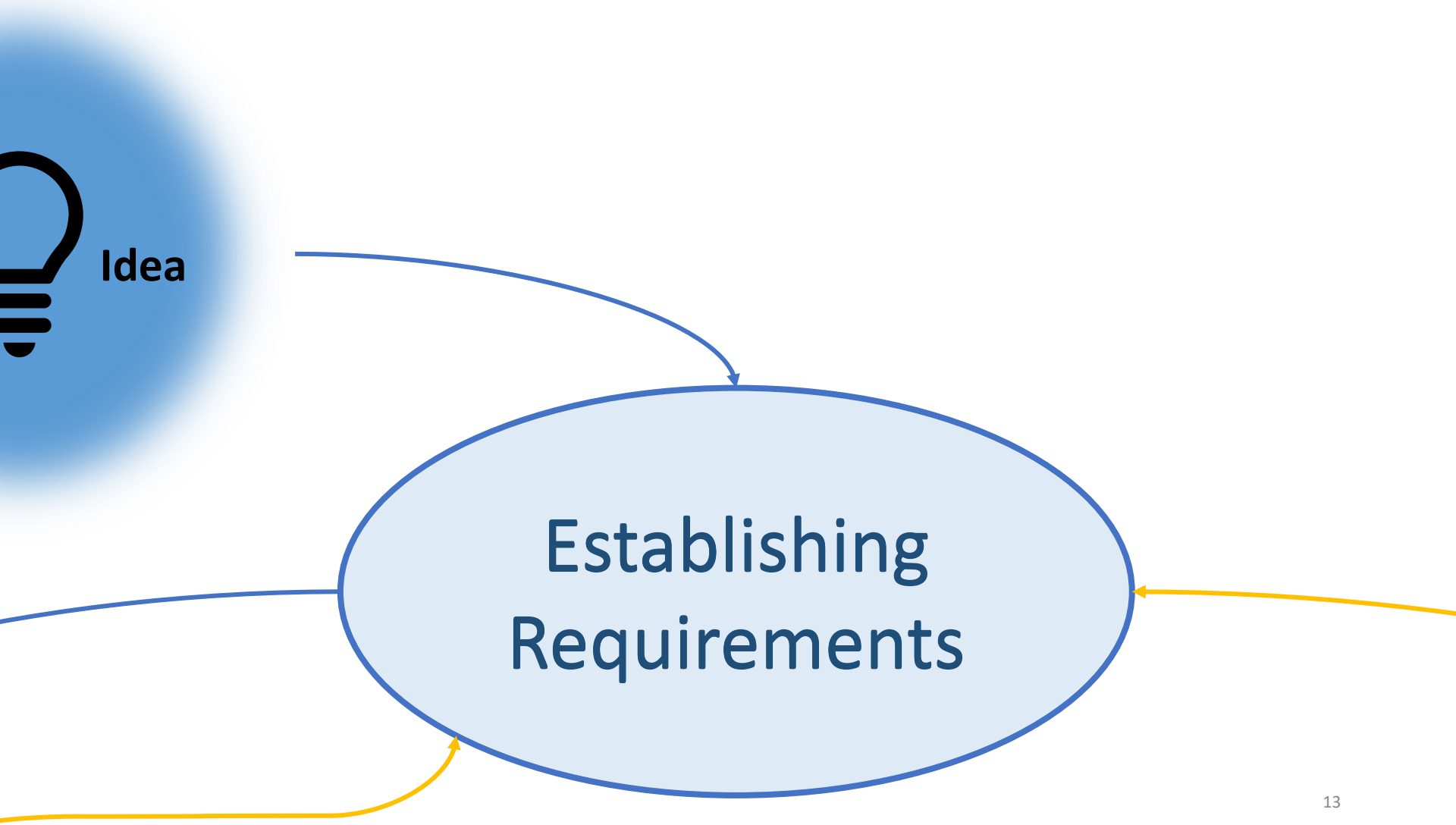


Design Process

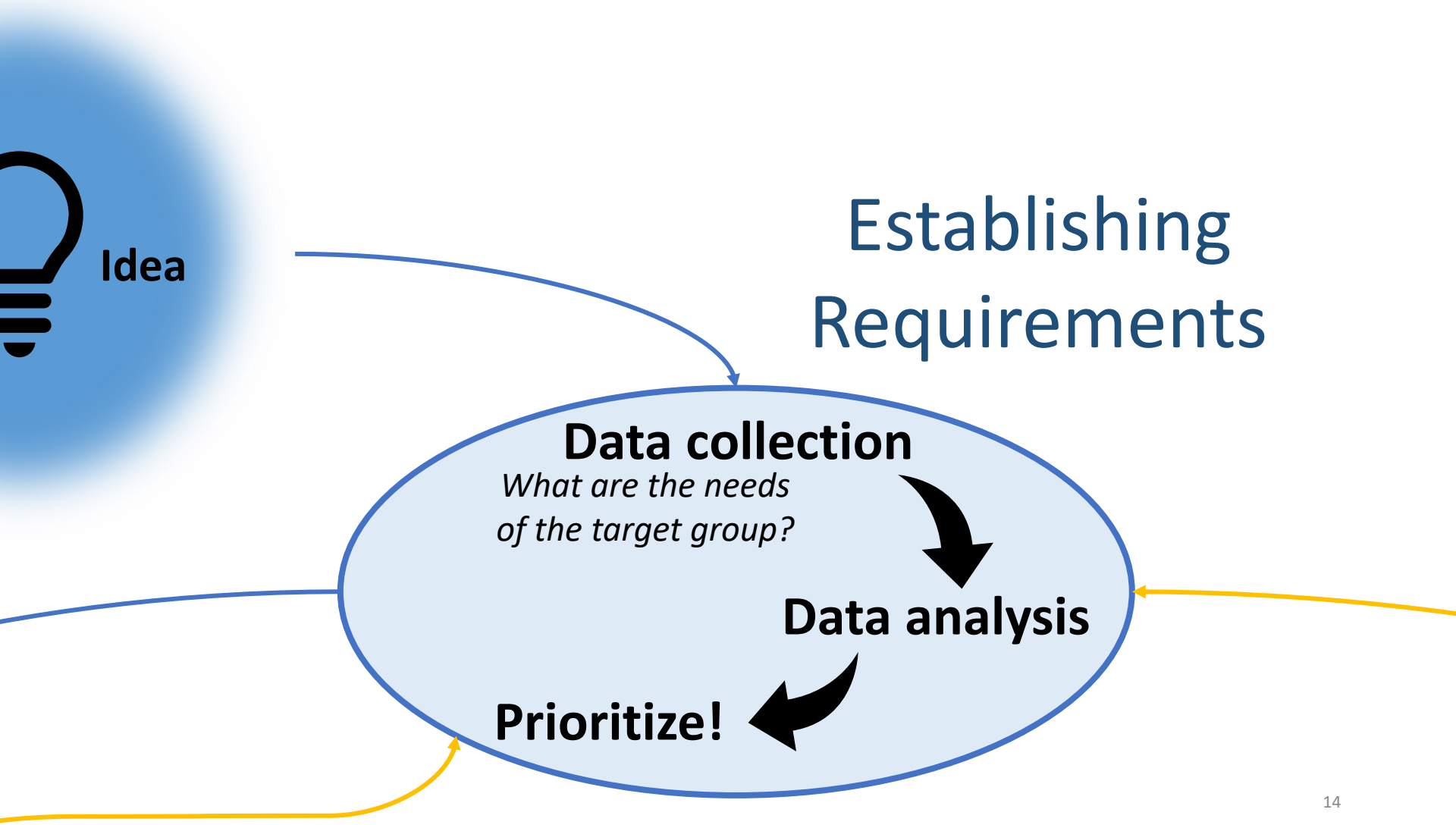


Preece, J., Sharp, H., & Rogers, Y. (2015). **Interaction design: beyond human-computer interaction**. John Wiley & Sons.

Design Process



Design Process

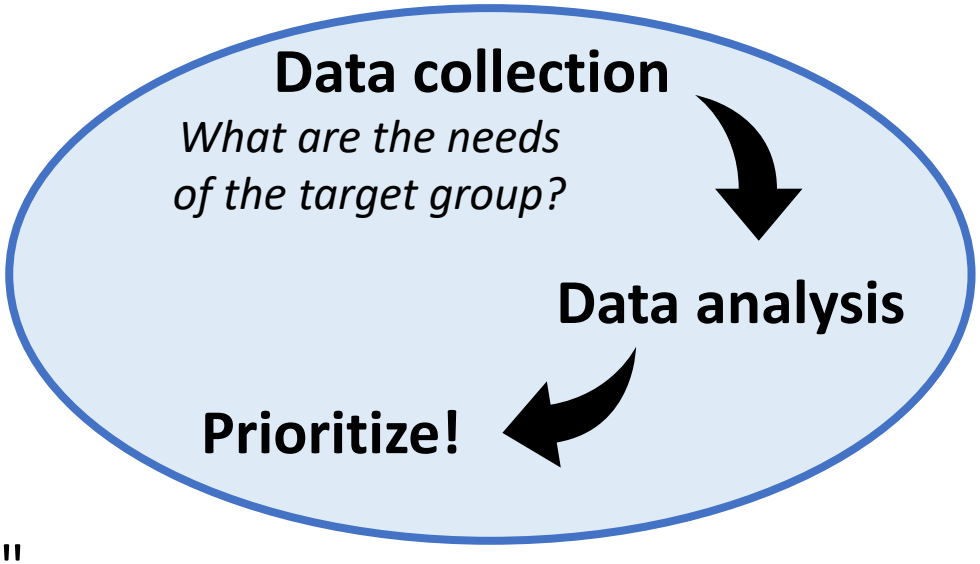


First Stage – Establish requirements

Find out if your target group has the need/problem you think

Through following steps:

- Data collection
- Data analysis
- Possibly update target group and/or needs/problems



- Produce a "requirements list" (prioritized list of features)

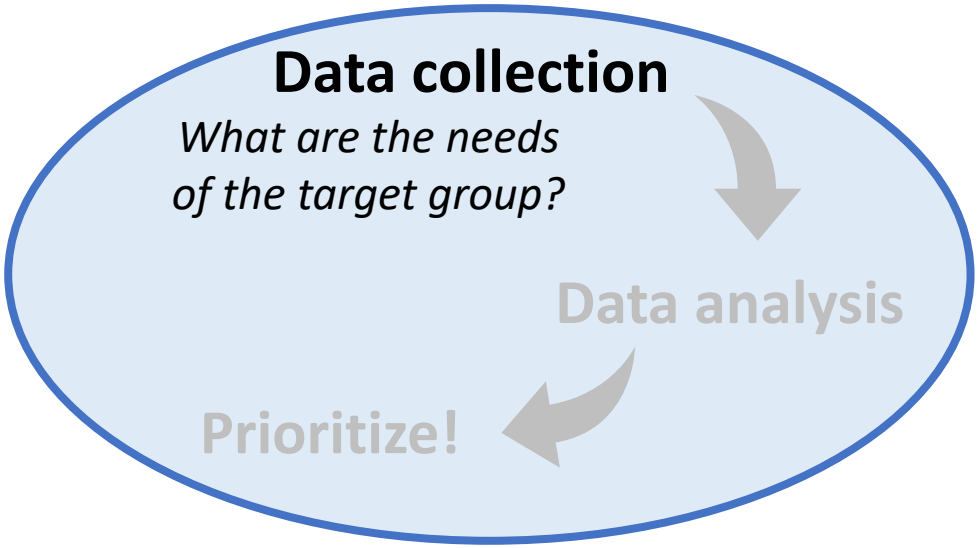
Set your goals!

Why do you want to collect “data”?

What is this “data” you seek?

Specify your goals

- Features
- Usability
- Look / feel / aesthetics
- Hierarchy



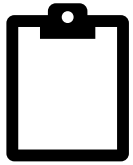
Identify the participants

Who is the target group?

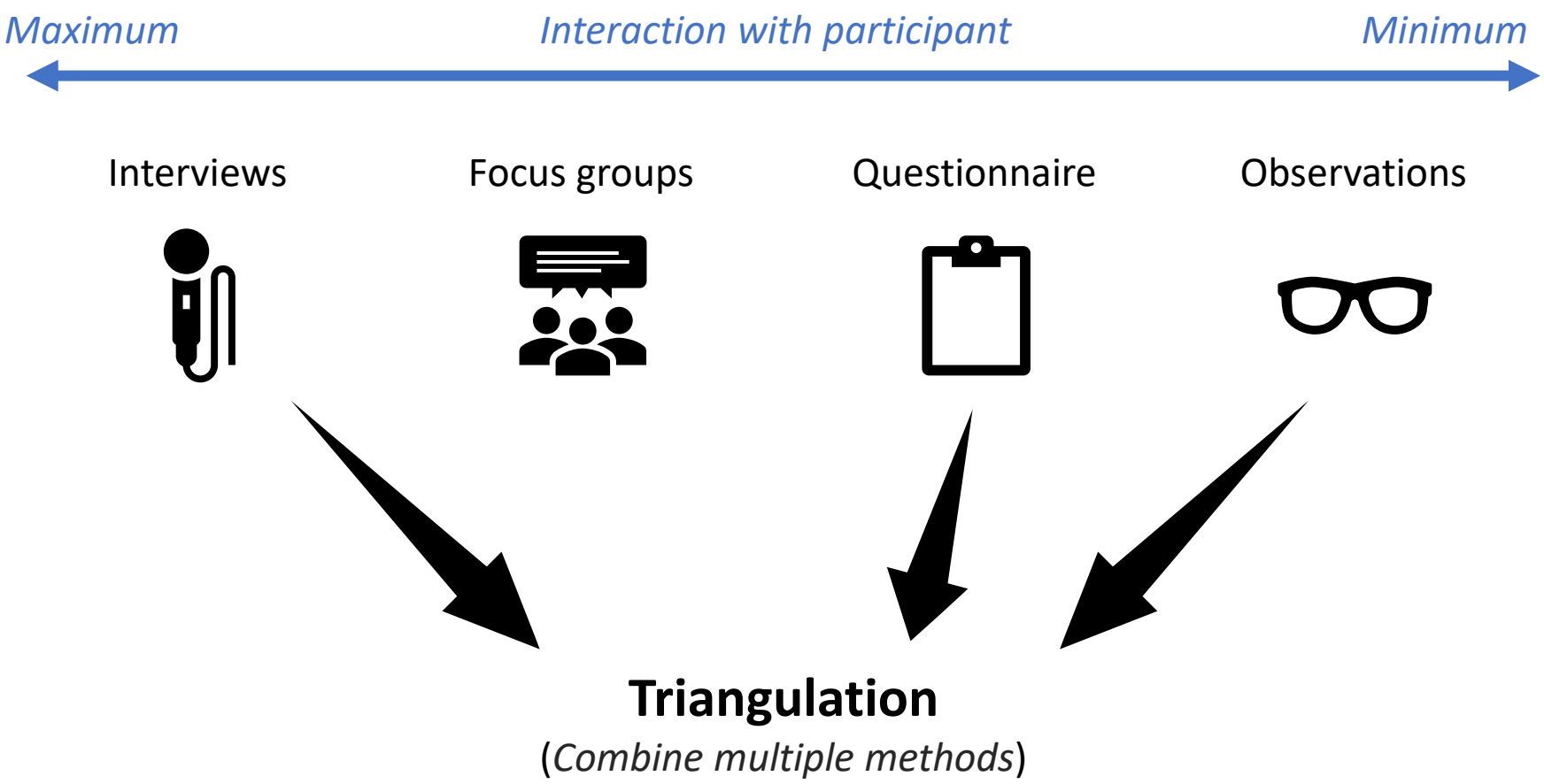
- Students?
- Elderly people?
- Workers at a factory?
- Drivers?

Data Collection Methods

Maximum — *Interaction with participant* — *Minimum*

Interviews	Focus groups	Questionnaire	Observations
			
Structured	Needs guidance	Pre-selected questions	Direct obs. in lab
Semi-structured	“*Groupthink*” !!	Open questions	Direct obs. in the wild
Unstructured			Indirect obs.

Data Collection Methods



Choosing techniques

Each techniques has advantages and disadvantages!

- What is the focus of the study?
- Who are the participants?
- What about technique – what can be done with it?
- Resources?

Data Collection



1. Define the problem to which you want answers



2 . Decide which method to use for your survey. Perhaps you can search for statistics, interview, observe or send out a survey.



3. Make a selection. Where, when, which, how many?



4. Collect data according to the method you chose.

Justify your choice of method(s) and selection in the report

Don't ask customers what they want .

Measure what they do.

What else to think?

There are certain important things to consider, especially when it comes to practicality.

What have I missed?

Exercise – Prepare for qualitative data collection

Think about:

- Place, time, target group
- Information for interviewees.
  - Presentation of you, how long it takes, how the data is used (anonymously?) etc. (Read about *informed consent form* in the book)
- Formulate qualitative questions and review them yourself
  - Is the question open?
  - Is the question clean and defined?
  - Is the question clear? (facilitates analysis)
  - Can the question be specified? (more interesting result)

  Example opening question: Can you tell us about an occasion when you experienced...? (broad, open question)

  Examples of deepening questions: How did it feel then?, What did you think then?, It was interesting, tell me more...
- Recording method - note or record
- Who interviews, who records